

- 1) Write a C program for developing a Lexical analyzer (LA) that will eliminate white spaces from a source program written in C and collect numbers.

input file link- <https://dpaste.com/2QAZEBUAI>

main code link- <https://dpaste.com/EPQBE5GLR>

- 2) Write a C program for developing a lexical analyzer (LA) that will recognize the keywords in a source program.

<https://dpaste.com/6YHRTXTD5>

- 3) Write a C program for developing a lexical analyzer (LA) that will recognize the variables in a source program.

<https://dpaste.com/EGRE95XT8>

- 4) Write a C program for developing a lexical analyzer (LA) that will find out the single as well as multi line comments.

<https://dpaste.com/9A4ZKBWUS>

1) Write a C program for developing a Lexical analyzer (LA) that will eliminate white spaces from a source program written in C and collect numbers.

```
#include<bits/stdc++.h>
#include<stdio.h>
#include<ctype.h>
using namespace std;

int main(){
    FILE *fp;
    int ch;
    fp=fopen("lex.cpp","r");
    if(fp==NULL){
        cout<<"File cannot open";
        return 0;
    }
    cout<<endl;

    while((ch=fgetc(fp))!=EOF){
        if(!isspace(ch)) putchar(ch);
    }
    rewind(fp);

    while((ch=fgetc(fp))!=EOF){
        if(isdigit(ch)){
            while(isdigit(ch)){
                putchar(ch);
                ch=fgetc(fp);
            }
            cout<<endl;
        }
    }
    fclose(fp);
}
```

- 2) Write a C program for developing a lexical analyzer (LA) that will recognize the keywords in a source program.

```
#include<bits/stdc++.h>
#include<fstream>
#include<string>
using namespace std;
int main(){
    string keyword[32]={
        "auto","break","case","char","continue","default",
        "do","double","else","enum","extern","float","for","goto",
        "if","int","long","register","return","short","signed",
        "sizeof","static","struct","switch","typedef","union",
        "unsigned","void","volatile","while"
    };

    ifstream file("lex.cpp");

    if(!file){
        cout<<"File not found!";
        return 0;
    }

    string word;

    while(file>>word){
        for(int i=0;i<32;i++){
            if(word==keyword[i]){
                cout<<word<<"-->Keyword"<<endl;
                break;
            }
        }
    }
    file.close();
}
```

Q: file >> word কী করে?

Ans: file >> word ফাইল থেকে একটি word (whitespace পর্যন্ত) পড়ে এবং সেটি word ভ্যারিয়েবলে সংরক্ষণ করে। প্রতিবার কল করলে পরবর্তী word পড়ে। File শেষ হলে read operation fail করে এবং while(file >> word) লুপ বন্ধ হয়ে যায়।

- 3) Write a C program for developing a lexical analyzer (LA) that will recognize the variables in a source program.

```
#include<bits/stdc++.h>
#include<fstream>
using namespace std;
int main()
{
    string keyword[32]={
        "auto","break","case","char","const","continue","default",
        "do","double","else","enum","extern","float","for","goto",
        "if","int","long","register","return","short","signed",
        "sizeof","static","struct","switch","typedef","union",
        "unsigned","void","volatile","while"
    };
    ifstream file("lex.cpp");
    if(!file)
    {
        cout<<"File not found!";
        return 0;
    }
    string word;
    while(file >> word)
    {
        bool isKeyword = false;
        for(int i=0; i<32; i++)
        {
            if(word == keyword[i])
            {
                isKeyword = true;
                break;
            }
        }
        if(!isKeyword && isalpha(word[0]))
        {
            cout << word << " --> Variable" << endl;
        }
    }
    file.close();
}
```

```
if(!isKeyword && isalpha(word[0]))
```

এখানে দুইটি শর্ত আছে:

- `!isKeyword` → Keyword হবে না।
- `isalpha(word[0])` → প্রথম অক্ষর অবশ্যই ইংরেজি বর্ণ (A-Z বা a-z) হতে হবে।

তাহলে সেটিকে Variable হিসেবে Print করবে।

- 4) Write a C program for developing a lexical analyzer (LA) that will find out the single as well as multi line comments.

```
#include<bits/stdc++.h>
#include<fstream>
using namespace std;
int main()
{
    ifstream file("lex.cpp");
    if(!file)
    {
        cout<<"File not found!";
        return 0;
    }
    string line;
    while(getline(file, line))
    {
        if(line.find("//") != string::npos)
        {
            cout<<"Single Line Comment : "<<line<<endl;
        }
        if(line.find("/*") != string::npos)
        {
            cout<<"Multi Line Comment Starts"<<endl;
        }
        if(line.find("*/") != string::npos)
        {
            cout<<"Multi Line Comment Ends"<<endl;
        }
    }
    file.close();
}
```

প্রথম অংশ

line.find("//")

মানে

"//" খুঁজো।

দ্বিতীয় অংশ

```
!= string::npos
```

মানে

"পাওয়া যায়নি" এর সমান না।

অর্থাৎ

পাওয়া গেছে।

তাই পুরো লাইনের বাংলা

```
if(line.find("//") != string::npos)
```

=

যদি লাইনে // পাওয়া যায় তাহলে...